Nazib Abrar

abrarnazib@gmail.com — github.com/abrar-nazib — linkedin.com/in/abrar-nazib

Education

Rajshahi University of Engineering & Technology March 2022 - Present
BSc in Mechatronics Engineering — Expected Graduation Date August 2026

Notre Dame College July 2018 - August 2020
HSC from Science — GPA 5.00

Work Experience

AI/ML Engineer (Contract) — DamDekho.com March 2026 - Present

- Architected and developed the full agentic crawling, data cleaning, and ingestion pipeline for a large-scale price-comparison catalog.
- Built an 8-stage AI normalization pipeline in **Python** using **LLMs** for structured field extraction and **vector embeddings** for semantic product matching against **PostgreSQL** + **pgvector**.
- Implemented a 3-tier **LLM** and embedding provider chain (GPU node over **WebSocket** → local CPU inference → cloud), with a multi-provider router supporting hot-swappable backends.
- Developed an autonomous **LLM**-driven agent in **Python** that reviews crawler configs, repairs catalog data, and approves normalized rows against a deterministic rubric via a typed **REST** API.

Embedded Systems Engineer (Contract) — Aethes Armor (USA) March 2025 - January 2026

- Engineered a Debian based custom embedded Linux distribution for a security locker device using the **Yocto Project**, including writing custom device drivers and patching the kernel.
- Developed the device's core firmware, implementing **WebRTC** for live video-calling and **Websockets** for real-time control of peripherals.
- Integrated computer vision and **AI** functionalities using **TensorFlow-Lite** to enable intelligent device operations.
- Built the complete web backend and server infrastructure for the system using **Django**.
- Contributed to the hardware development lifecycle by assisting in **PCB** design reviews and Bill of Materials (BOM) generation.

Junior Software Developer (Part-Time) — FronTech Limited June 2023 - October 2025

- Created well-documented libraries in **C++** for JRC Board (an ESP32-based microprocessor development board) to ensure compatibility with Arduino shields.
- Led back-end development of a scalable alumni management system using **Django Rest Framework** and **PostgreSQL**.
- Integrated **RabbitMQ** and **Celery** to handle asynchronous tasks, enabling efficient processing of time-consuming operations. Deployed the project in a microservice architecture using **Docker**, emphasizing scalability and seamless deployment.
- Led the back-end development for the "smartschoolbus.gov.bd" project, a B2B solution built with **NodeJS**, **Express**, and **MongoDB** to improve school bus safety and efficiency for over 2,500 active users.
- Managed **VPS** infrastructure, utilizing **Nginx** for traffic routing and access control, and implemented **CRON** jobs for automated database backups.
- Implemented live video streaming in **RTMP** and **HLS** protocol.

Research & Projects

StereoLite: A Novel Stereo Vision Architecture for Disparity Matching and Depth Estimation

- Designed a novel lightweight stereo matching architecture that fuses a tile-hypothesis cost-volume paradigm with an iterative refinement loop and learned convex upsampling, targeting real-time depth estimation on edge hardware.
- Conducted a systematic literature review of ∼190 deep stereo matching papers and authored a comparative analysis spanning classical SGM to foundation-model-era methods, informing the architecture's design choices.
- Ran a multi-phase ablation study (9 architectural variants A/B-tested in parallel on Modal A100s) covering cost-volume designs, slope-aware warping, context branches, and several loss-side interventions to isolate the contributing components.
- Implemented the model in **PyTorch** with a custom multi-scale tile refinement loop and groupwise 1D cost volume, benchmarked against published lightweight stereo baselines on Scene Flow and indoor datasets.

CORTEX-Health: An AI Assistant for Medical Practitioners (PyTorch, FastAPI, OpenCV)

- Developed a **FastAPI**-based API server to facilitate communication between an Android application (built with Flutter) and machine learning models. Used **PostgreSQL** as the database.
- Trained three YOLO-v8 models using **PyTorch** for disease diagnosis from medical images (e.g., X-RAY and CT scan reports).

Leaderships and Awards

Secretary of Programming Dept — Notre Dame Information Technology Club

- Led programming-related affairs, including contests and tutoring sessions for junior members.

Champion - Bangladesh Round, 4^{th} Global - 6^{th} Kibo RPC 2024

- Achieved 1^{st} position in the Bangladesh Round and 4^{th} position globally in the Kibo RPC competition, developing software to control a free-flying robot in zero gravity. Employed computer vision techniques for navigation and task execution.

Global 11^{th} Position – International Rover Design Challenge 2023

- Contributed to Team Ogrodoot's best performance on IRDC by implementing computer vision based mars terrain segmentation system using SAM and designing inverse kinematics algorithm for robotic arm of the rover.

First Runner-Up - Phitron Show Your Project Contest 2022

- Won the project showcase competition with CORTEX Robotic Arm Controller Software.

Champion - DRMC Tech Carnival 2020

- Won the Line Follower Robot racing competition using "Thunder," an LFR controller software developed with C++.

Finalist - Code Samurai 2024 (Hackathon)

- Advanced to the final round in a competitive field of 400+ teams, achieving 14th place in the Code Samurai 2024 Hackathon with an end-to-end solution for municipal solid waste management.

Problem-Solving Experience

- Solved 85+ problems on LeetCode following the Neetcode.io roadmap, gaining a strong understanding of diverse data structures and algorithms. [Profile Link]
- Codechef Rating 1550. [Profile Link]

Certificates

Machine Learning Specialization September 2023
Issued by: Stanford Online